

Contractor's Root Cause Analysis — Pier P2 Pier Cap Cracking

Nandira River Bridge Project — Package NRB-4

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From:	Contractor, Quality Manager, Sagara Constructions Pvt. Ltd.
To:	Engineer, Meridian Engineering Consultants

Subject: Root Cause Analysis — Pier P2 Pier Cap Cracking

Further to Defect Notice ENG-NRB4-0078, and having reviewed Crack Mapping Report CMR-NRB4-001, Laboratory Test Report LAB-NRB4-058, and our own pour record and Site diary for the pour of 11-Aug-2021, we report the following root cause analysis.

1. WEATHER RECORDS: Site weather records for 11-Aug-2021 show a maximum ambient temperature of 33.5 degrees C, within the range recorded at other substructure pours completed successfully during the same period, including the Pier P4 and Pier P5 pier caps (32-35 degrees C range, per pour records reviewed). We do not consider ambient temperature on the day of the pour to have been exceptional or a material contributing cause.
2. MIX DESIGN AND MATERIALS: The pour used the same approved Grade M40 mix design, from the same approved concrete source, as used across all other substructure elements on the Works, including Piers P1, P4 and P5, none of which exhibits comparable cracking. Petrographic examination in LAB-NRB4-058 found no abnormality in the mix. We do not consider the mix design or materials to be a contributing cause.
3. CURING RECORDS: Review of the Site diary for this pour identifies that the curing membrane and wet hessian cover specified for the exposed top surface were removed on 14-Aug-2021, approximately 3 days after the pour, to allow formwork for the bearing plinths to proceed ahead of programme, against a 7-day minimum curing period for exposed surfaces under the Specification. We have not identified a corresponding curing record for any other substructure pour on the Works showing an equivalent shortfall.
4. CONCLUSION: Having reviewed the alternative causes, we consider it more likely than not that the early removal of curing protection identified in paragraph 3 above is the principal contributing cause of the cracking, resulting in accelerated early-age drying shrinkage of the exposed top surface before the concrete had gained sufficient tensile strength to resist the resulting stresses. We accept this is a departure from the curing period required by the Specification.

Submitted by: Quality Manager, Sagara Constructions Pvt. Ltd.